

A Dual Mandate Can Support Price Stability

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¹Respectively, FRB Kansas City and FRB San Francisco. These views are those of the authors and do not necessarily reflect the views of the Federal Reserve Banks of Kansas City, San Francisco or the Federal Reserve System.

Motivation - Does a dual mandate hinder achieving price stability?

Central Bank mandated goals:

- Central Banks typically tasked with achieving price stability
- In some instances also given a goal of promoting maximum employment

Recurring question for policy makers: does a dual mandate hinder achieving price stability?

- A strong labor market fuels inflation via the Philips curve, the steeper the curve, the greater the challenge

Preview - Employment goal imparts history dependence in policy

(Un)Employment dynamics are persistent

- Strong empirical regularity in all advanced economies
- Feature in models of frictional labor markets: unemployment is a backward-looking stock
 - Unemployment **depends on the full history of past matches**

Behavior of monetary policy under a dual mandate central bank will be history dependent

- A DM thus effectively stabilizes inflation to demand shock
- Policymakers stabilizing the labor market will produce outcomes similar to AIT
- Plays an important stabilizing feature *at and away from* the zero lower bound

Preview - Key results

Analytical results in 3-equation NK framework

- A dual mandate reduces inflation volatility following demand shocks
- Derive conditions for observational equivalence with Average Inflation Targeting
- Supply shocks break this observational equivalence:
 - A dual mandate tolerates more inflation to stabilize real activity

Quantitative NK model with search frictional unemployment

- A dual mandate reduces inflation volatility by 26 percent relative to single inflation targeting
- Inflation IRFs to:
 - demand shocks identical under DM and AIT
 - supply shocks: dampened under AIT compared to a DM

Related literature

Monetary policy goals and strategies:

Blanchard and Gali (2010) and Ravenna and Walsh (2011); Altig et al. (2020);

Monetary policy at the zero lower bound

Eggertsson and Woodford (2003); Kiley and Roberts (2017); Mertens and Williams (2019); Amano et al. (2020); Nessén and Vestin (2005); Albertini and Poirier (2015)

Inflation and labor market dynamics

Walsh (2005), Krause and Lubik (2007), Gertler, Sala and Trigari (2008), Christiano, Eichenbaum and Trabandt (2016)

Roadmap

- 1 - Overview
- 2 - **Persistent unemployment in the data**
- 3 - Illustrative simple model
- 4 - Quantitative results: NK model with equilibrium unemployment

Unemployment growth has momentum across countries

Autocorrelation of quarterly unemployment growth

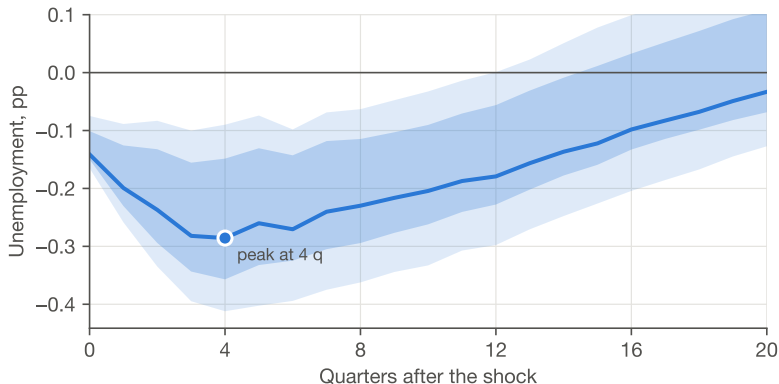
	1995–2019		Full sample	
	Lag 1	Lag 2	Lag 1	Lag 2
Data:				
United States	0.67	0.59	0.64	0.30
United Kingdom	0.47	0.41	0.46	0.40
Canada	0.40	0.21	0.54	0.32
Germany*	0.71	0.52	0.75	0.55
Italy	–	–	0.64	0.55

Notes: National seasonally adjusted rates; samples end 2019Q4. Full samples start in 1949 (US), 1976 (Canada), 1992 (UK, Germany) and 2004 (Italy).

*Registered unemployment.

Unemployment responds to demand shocks with a hump

Response of unemployment to an expansionary demand shock



Notes: SVAR in productivity growth and the unemployment gap, 1995Q1–2019Q4, 8 lags; one-standard-deviation shock; 68 and 90 percent bootstrap bands.

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Simple 4 period NK Model

Analytically highlight the history-dependence introduced by the dual mandate

New-Keynesian IS Curve:

$$x_t = E_t x_{t+1} - \left(i_t - E_t \pi_{t+1} - r_t^n \right)$$

Forward-looking Phillips Curve

$$\pi_t = \beta E_t \pi_{t+1} + \kappa x_t$$

x_t : Deviations of output from its potential

π_t : Inflation in deviations from zero inflation steady state

i_t : Central bank's short-term policy rate

r_t^n : Economy's natural rate of interest (affected by many shocks)

Simple 4 period NK Model - Three monetary policy rules

1 Strict inflation targeting:

$$i_t = \phi_\pi \pi_t$$

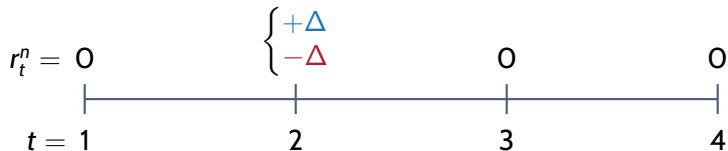
2 Dual mandate:

$$i_t = \phi_\pi \pi_t + \phi_u u_t$$

3 Average Inflation Targeting

$$i_t = \phi_\pi \pi_t + \phi_\pi^a \pi_{t-1} \quad \text{with} \quad 0 < \phi_\pi^a < \phi_\pi$$

Simple 4 period NK Model - Timeline



Simplified process for the natural rate r_t^n :

- Positive or negative Δ with equal probability in $t = 2$, 0 otherwise

Outcomes under strict inflation targeting: $i_t = \phi_\pi \pi_t$

		$\begin{cases} +\Delta \\ -\Delta \end{cases}$		
$r_t^n = 0$			0	0
$t = 1$				
			3	4
$x_t = 0$		Δ		
		$1 + \phi_\pi \kappa$	0	0
$\pi_t = 0$		$\kappa \Delta$		
		$1 + \phi_\pi \kappa$	0	0

Solving backwards

- Gaps at 0 in $t = 4$ and 3; symmetry of outcomes in $t = 2$ that $E_1 x_2$ and $E_1 \pi_2 = 0$
- Policy plays a key role in stabilizing economy when hit with shocks (ϕ_π)

Outcomes under strict inflation targeting: $i_t = \phi_\pi \pi_t$

$r_t^n = 0$	$\begin{cases} +\Delta \\ -\Delta \end{cases}$	0	0
$t = 1$	2	3	4
$x_t = 0$	$\frac{\Delta}{1 + \phi_\pi \kappa}$	0	0
$\pi_t = 0$	$\frac{\kappa \Delta}{1 + \phi_\pi \kappa}$	0	0

Solving backwards

- Gaps at 0 in $t = 4$ and 3; symmetry of outcomes in $t = 2$ that $E_1 x_2$ and $E_1 \pi_2 = 0$
- Policy plays a key role in stabilizing economy when hit with shocks (ϕ_π)

Dual mandate under persistent unemployment gaps

Assumptions on the unemployment gap:

- 1 - matches out of unemployment are proportional to economic activity
- 2 - one period matches for analytical tractability

$$u_t = (1 - s)u_{t-1} - \frac{1}{c}x_{t-1} \quad \rightarrow \quad u_t = -\frac{1}{c}x_{t-1}$$

Dual mandate under persistent unemployment gaps

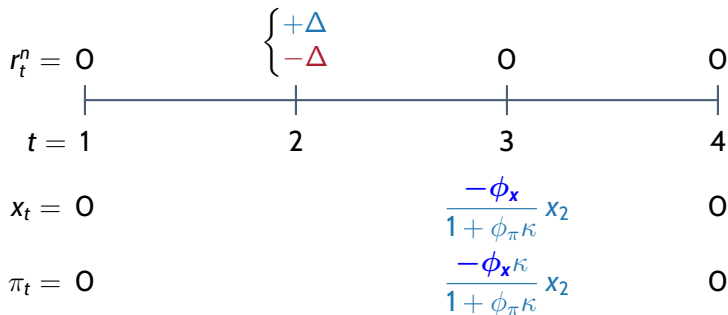
Central bank now follows a dual mandate (in $t = 3$): $i_t = \phi_\pi \pi_t + \phi_u u_t$

- Expressed as a function of the output gap:

$$i_t = \phi_\pi \pi_t + \phi_x x_{t-1} \quad \text{where} \quad \phi_x = -\frac{\phi_u}{c}$$

⇒ when the evolution of the labor market depends on past outcomes, the behavior of monetary policy also becomes history dependent

Impact of a dual mandate: $i_3 = \phi_\pi \pi_3 + \phi_x x_2$



Solving backwards: $t = 3 \Rightarrow$ Central bank offsets $t = 2$ fluctuations in x the following period

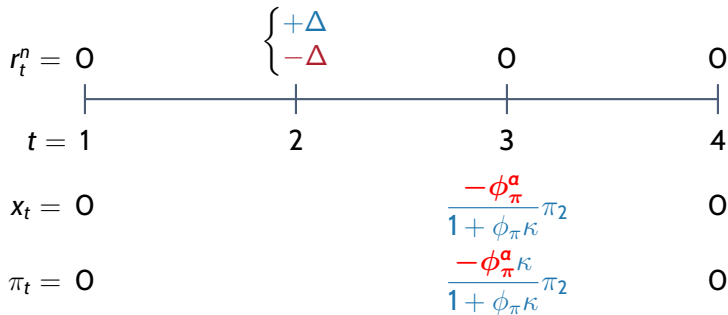
Impact of a dual mandate: $i_3 = \phi_\pi \pi_3 + \phi_x x_2$

$r_t^n = 0$		$\begin{cases} +\Delta \\ -\Delta \end{cases}$	0	0
$t = 1$		2	3	4
$x_t = 0$		$\tilde{\psi}^D \times \frac{\Delta}{1 + \phi_\pi \kappa}$	$\frac{-\phi_x}{1 + \phi_\pi \kappa} x_2$	0
$\pi_t = 0$		$\psi^D \times \kappa x_2$	$\frac{-\phi_x \kappa}{1 + \phi_\pi \kappa} x_2$	0

Key implication: this future response is anticipated

- dampening the response of real activity: $\tilde{\psi}^D \equiv \frac{1 + \phi_\pi \kappa}{1 + \psi^D \phi_\pi \kappa + \phi_x \frac{1 + \kappa}{1 + \phi_\pi \kappa}} < 1$
- flattening of the reduced-form Phillips curve: $\psi^D \equiv \left(1 - \frac{\phi_x \beta}{1 + \phi_\pi \kappa}\right) < 1$ if $\phi_x > 0$

Impact of AIT: $i_3 = \phi_\pi \pi_3 + \phi_\pi^a \pi_2$



Solving backwards: $t = 3 \Rightarrow$ Central bank offsets $t = 2$ fluctuations in π the following period

Impact of AIT: $i_3 = \phi_\pi \pi_3 + \phi_\pi^a \pi_2$

		$\begin{cases} +\Delta \\ -\Delta \end{cases}$	0	0
$r_t^n = 0$				
$t = 1$	1	2	3	4
$x_t = 0$		$\tilde{\psi}^{AIT} \times \frac{\Delta}{1 + \phi_\pi \kappa}$	$\frac{-\phi_\pi^a}{1 + \phi_\pi \kappa} \pi_2$	0
$\pi_t = 0$		$\psi^{AIT} \times \kappa x_2$	$\frac{-\phi_\pi^a \kappa}{1 + \phi_\pi \kappa} \pi_2$	0

Key implication: this future response is anticipated

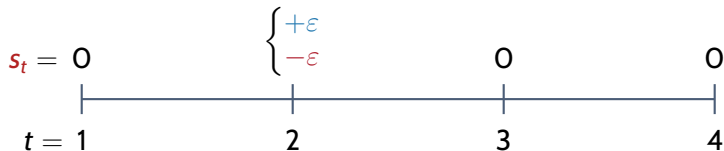
- dampening the response of real activity: $\tilde{\psi}^{AIT} \equiv \frac{1 + \phi_\pi \kappa}{1 + \psi^{AIT} \kappa \left(\phi_\pi + \phi_\pi^a \frac{1 + \kappa}{1 + \phi_\pi \kappa} \right)} < 1$
- flattening of the reduced-form Phillips curve: $\psi^{AIT} \equiv \frac{1 + \phi_\pi \kappa}{1 + \phi_\pi \kappa + \beta \phi_\pi^a \kappa} < 1$ if $\phi_\pi^a > 0$

Dual mandate was already providing benefits similar to AIT

	Phillips Curve Slope	Output Gap x_2	
		Away from ZLB	At ZLB
Single Mandate	κ	$-\frac{\Delta}{1 + \phi_\pi \kappa}$	$-\Delta^{ZLB}$
Dual Mandate	$\psi^D \kappa$	$-\tilde{\psi}^D \frac{\Delta}{1 + \phi_\pi \kappa}$	$-\phi^D \Delta^{ZLB}$
AIT	$\psi^{AIT} \kappa$	$-\tilde{\psi}^{AIT} \frac{\Delta}{1 + \phi_\pi \kappa}$	$-\phi^{AIT} \Delta^{ZLB}$

Notes: AIT denotes average inflation targeting. Columns 2 and 3 report the output gap in period 2 when the natural rate $r_2^n = -\Delta$ (column 2) or $r_2^n = -i - \Delta^{ZLB}$ (column 3). Each colored term is less than one, implying stabilization relative to the single mandate:

Supply shocks break the observational equivalence

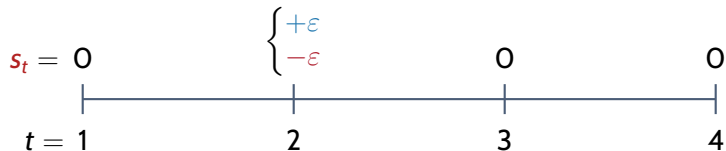


Consider an inefficient supply shock to the Phillips curve ($r_t^n = 0$ at all time):

$$\pi_t = \beta E_t \pi_{t+1} + \kappa x_t + s_t,$$

- s_t : Positive or negative ε with equal probability in $t = 2$, 0 otherwise
- Creates a trade-off: inflation rises while output falls

Dual mandate tolerates more inflation to stabilize real activity



Single mandate	π_2^{IT}	$\frac{1}{1 + \phi_\pi \kappa} \varepsilon$	
Dual mandate	π_2^D	$\gamma^D \times \frac{1}{1 + \phi_\pi \kappa} \varepsilon$	where $\gamma^D > 1$
AIT	π_2^{AIT}	$\gamma^{AIT} \times \frac{1}{1 + \phi_\pi \kappa} \varepsilon$	where $\gamma^{AIT} < 1$

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Model: key features

New Keynesian model with nominal rigidities, frictional labor market, and the zero lower bound:

- Representative large household supplying labor on extensive and intensive margins
- Monopolistically competitive retail firms face nominal price adjust costs
- Wholesale firms adjust labor inputs on employment and hours margins
 - Hire on a search frictional labor market
 - Bargain over hours and compensation
- Business cycles generated by two types of shocks:
 - Productivity shocks (to retail firms): positive shocks cause unemployment and inflation to fall
 - Demand shocks (to consumer preferences): positive shocks cause unemployment to fall and inflation to rise

[Parameters values] [Full model]

Empirical and Model-Implied Moments

	Data	Dual mandate	Strict inflation target	
			IT	AIT
Mean:				
U	5.7	5.1	5.1	5.1
π	1.8	1.8	1.6	1.7
R	2.5	2.4	2.2	2.3
Standard deviation:				
$\sigma(U)$	0.11	0.07	0.07	0.07
$\sigma(\pi)$	0.44	0.40	0.54	0.40
$\sigma(R)$	0.39	0.63	0.79	0.62
Correlations:				
$\text{corr}(U, \pi)$	-0.44	-0.45	-0.61	-0.64
$\text{corr}(g_t, g_{t-1})$	0.67	0.28	0.29	0.28

Notes: $g_t = (U_t - U_{t-1})/U_{t-1}$, quarterly, 1995–2019. Model: mean of 1,000 samples; π^m : monthly inflation rate, annualized.

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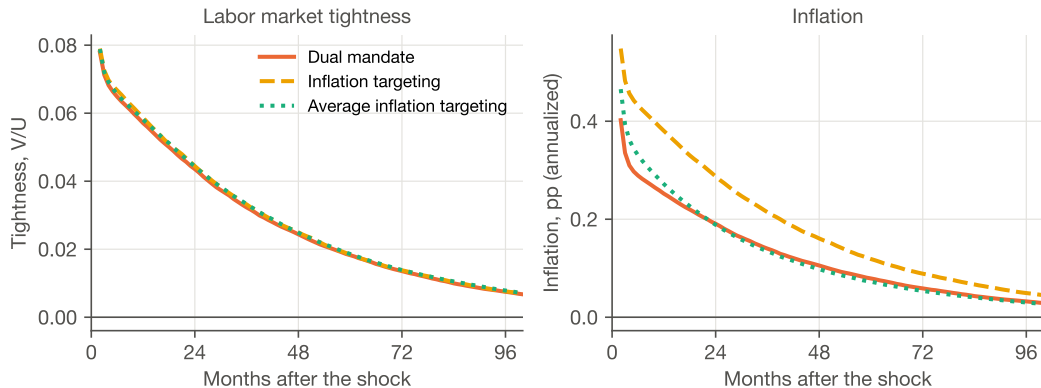
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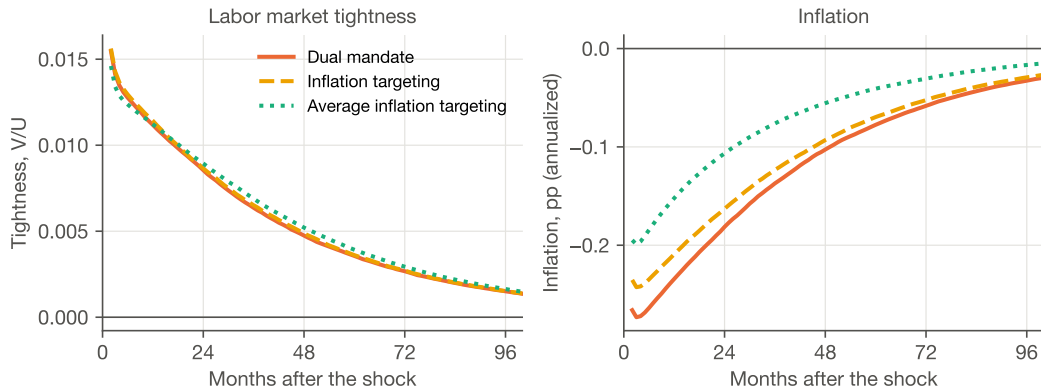
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Responses to Expansionary Demand Shocks



Notes: One-standard-deviation shock, starting from the steady state.

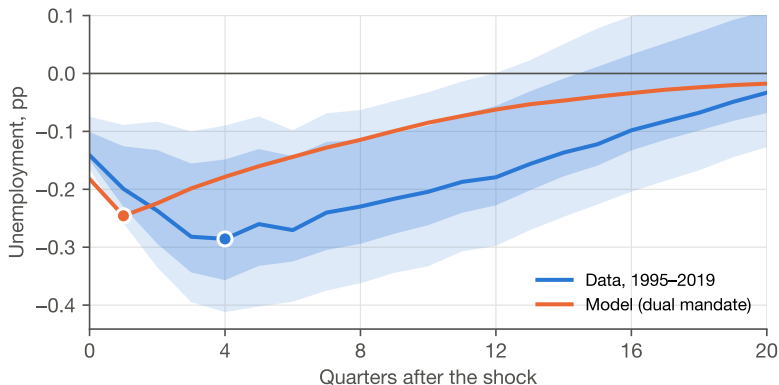
Responses to Expansionary Supply Shocks



Notes: One-standard-deviation shock, starting from the steady state.

Shortcomings: data and model responses

Response of unemployment to an expansionary demand shock



Notes: SVAR in productivity growth and the unemployment gap, 1995Q1-2019Q4, 8 lags; one-standard-deviation shock; 68 and 90 percent bootstrap bands. Model: median across 2,300 model samples of 100 quarters.

The Dual Mandate Can Support Price Stability

Additional slides

Stabilizing the economy at the zero lower bound



Introduce a large negative decline in demand that takes the economy to the ZLB with certainty

$$\Rightarrow r_2^n = -i - \Delta^{ZLB} \text{ where } \Delta^{ZLB} > 0$$

Outcomes under a single mandate: inability to offset the negative shock results in a large negative output gap and below-target inflation

- Policy rate: $i_t = \max(\phi_\pi \pi_t, -i)$
- Period 2 outcomes:

$$x_2^{-\Delta^{ZLB}} = -\Delta^{ZLB}, \quad \pi_2^{-\Delta^{ZLB}} = -\kappa \Delta^{ZLB}$$

Stabilizing the economy at the zero lower bound

Dual mandate policy rate: $i_t = \max \left(\phi_\pi \pi_t + \phi_u u_t, -i \right) \Rightarrow i_t = \max \left(\phi_\pi \pi_t + \phi_x x_{t-1}, -i \right)$

- $t = 3$ outcomes same as in earlier case
- $t = 2$ outcomes:

$$x_2 = \Psi \left(-\Delta^{ZLB} \right) \quad , \quad \pi_2 = \left(1 - \frac{\beta \phi_x}{1 + \phi_\pi \kappa} \right) \Psi \left(-\Delta^{ZLB} \right) \quad \text{where} \quad \Psi = \frac{1}{1 + \frac{\phi_x(1+\kappa)}{1+\phi_\pi \kappa}} < 1$$

Expectations of more accommodative policy in $t = 3$ lead to better outcomes during the downturn:

- Stabilizing term Ψ : more pronounced as the Central Bank places additional weight on stabilizing real activity (ϕ_x) relative to inflation (ϕ_π)
- Stabilization equivalence with AIT if $\phi_x = \phi_\pi^a \kappa$

Model: calibration (1/2)

Parameter	Notation	Value	Target/Source
Preferences:			
Discount factor	β	$1/(1 + e^{\frac{-0.5}{1200}})$	0.5% annualized real risk free rate
Elast. of subst. btw. goods	ω	10	Average markup over marginal cost
Utility: consumption	σ	2	External
Utility: leisure, level	ν_0	6.4	Average hours worked
Utility: leisure, curvature	ν_1	2	External
Non-employment utility	ν_u	-6.7	Reservation to wage ratio
Labor market:			
Matching function: curvature	ν	1.25	Den Haan et al (2000)
Worker bargaining weight	η	0.25	Unemployment rate
Vacancy cost	κ_0	0.04	Recruiting costs to monthly wage
Fixed hiring cost	κ_1	0.06	Volatility of unemployment
Job destruction rate	s	0.03	Unemployment flow accounting
Unemployment benefits	b	0.15	Replacement rate $\times$ coverage ratio

Model: calibration (2/2)

Parameter	Notation	Value	Target/Source
Price setting:			
Price adjustment cost	Ω	500	Volatility of inflation
Monetary policy:			
Weight on inflation	ϕ_{π}	1.50	Estimated on U.S. data
Weight on unemp. gap	ϕ_u	-0.04	Estimated on U.S. data
Inflation target	π^*	1.02 ^(1/12)	FOMC target inflation
Unemployment: natural rate	U^*	0.05	Long run rate of unemployment
AIT weight	ϕ_{π^a}	0.25	Volatility of inflation under dual mandate
AIT look-back	K	15	Approx. 1.25-year look-back at monthly freq.
Shock processes:			
Technology: persistence	ρ_x	0.98	U.S. labor productivity
Technology: standard deviation	σ_x	0.004	U.S. labor productivity
Demand: persistence	ρ_{γ}	0.98	Ireland (2011)
Demand: standard deviation	σ_{γ}	0.03	$\text{corr}(U, \pi)$

Model: Labor market

- Unemployed (U) find available jobs (V) at a rate governed by a CRS matching function $G(U, V)$
 - Tightness of the labor market summarized by the ratio $\theta = V/U$
 - Unemployed find a job at rate $\frac{G(U_t, V_t)}{U_t} = f(\theta_t)$ with $f'(\theta_t) > 0$
 - Job openings are filled at a rate $\frac{G(U_t, V_t)}{V_t} = q(\theta_t)$ with $q'(\theta_t) < 0$
- Employment loses their job at a constant rate s
- Employment and unemployment evolve according to:

$$N_{t+1} = (1 - s)N_t + q(\theta_t)V_t$$

$$U_{t+1} = (1 - f(\theta_t))U_t + sN_t.$$

Model: Households

Household value function taking share of employed and unemployed members as given:

$$J_t = \max_{C_t, B_t} \left\{ \exp(\gamma_t) \frac{C_t^{1-\sigma}}{1-\sigma} + \nu_0 \frac{(1-H_t)^{1-\nu_1}}{1-\nu_1} N_t + \nu_u U_t + \beta E_t[J_{t+1}] \right\}$$

subject to $C_t + T_t + \frac{B_t}{P_t R_t^N} = \frac{B_{t-1}}{P_t} + W_t H_t N_t + b U_t + D_t,$

and l.o.m. for unemployed and employed

- Wage W and hours worked on the job H determined by bargaining with the firm
- Preference shifter: $\gamma_t = \rho_\gamma \gamma_{t-1} + \sigma_\gamma \varepsilon_t^\gamma,$

Model: Retail sector

Monopolistically competitive retail firms facing nominal price adjust costs:

$$S'_t \equiv \max_{P_t(j)} E_t \left[\sum_{i=0}^{\infty} M_{t,t+i} \left[\left(\frac{P_{t+i}(j)}{P_{t+i}} - \psi_{t+i} \right) Y_{t+i}^d(j) - \frac{\Omega}{2} \left(\frac{P_{t+i}(j)}{\Pi P_{t+i-1}} - 1 \right)^2 Y_{t+i} \right] \right] \quad (1)$$

- ψ_t : input of intermediate goods used by retailer j , purchased on a competitive intermediate goods market

The optimal price setting decision implies that, in a symmetric equilibrium with $P_t(j) = P_t$ for all j is:

$$\frac{\Pi_t}{\Pi} \left(\frac{\Pi_t}{\Pi} - 1 \right) = \frac{\omega}{\Omega} \left(\psi_t - \frac{\omega - 1}{\omega} \right) + E_t M_{t,t+1} \frac{Y_{t+1}}{Y_t} \frac{\Pi_{t+1}}{\Pi} \left(\frac{\Pi_{t+1}}{\Pi} - 1 \right) \quad (2)$$

Model: Wholesale sector

Whole firm adjust production by varying employment and hours of work:

$$S_t^w \equiv \max_{\{V_{t+i}, N_{t+i+1}\}_{i=0}^{\infty}} E_t \left[\sum_{i=0}^{\infty} M_{t,t+i} [\psi_{t+i} X_{t+i} N_{t+i+1} H_{t+i}^{\alpha} - W_{t+i} H_{t+i} N_{t+i} - \kappa_{t+i} V_{t+i}] \right]$$

- Production is linear in employment and concave in hours on worked on the job ($\alpha \in (0, 1)$) and TFP follows $\log(X_t) = \rho_x \log(X_{t-1}) + \sigma_x \varepsilon_t^x$,
- Job openings incur a per period cost $\kappa_t = \kappa_0 + \kappa_1 q_t > 0$

Optimal job posting define a job creation condition:

$$\frac{\kappa_t}{q(\theta_t)} - \lambda_t^V = E_t \left[M_{t,t+1} \left[\psi_{t+1} X_{t+1} H_{t+1}^{\alpha} - W_{t+1} H_{t+1} + (1-s) \left[\frac{\kappa_{t+1}}{q(\theta_{t+1})} - \lambda_{t+1}^V \right] \right] \right]$$

Model: Wages and hours of work

Firms and worker (Nash) bargaining over hours worked and compensation, solving the problem:

$$\Lambda_t = \max_{w_t, H_t} \left(\frac{J_{Nt} - J_{Ut}}{\lambda_t^C} \right)^\eta (S_{N,t}^w - S_{V,t}^w)^{1-\eta}$$

- Equilibrium hours condition equation MRS to MPL:

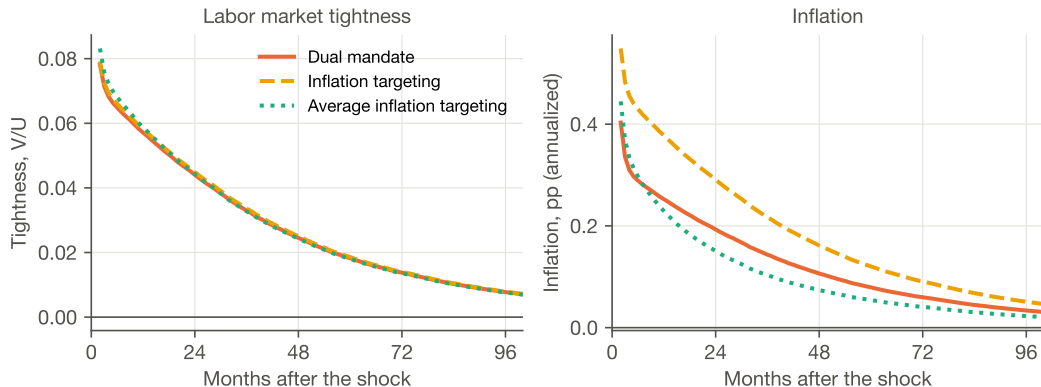
$$\frac{\nu_0}{\lambda_t^C} (1 - H_t)^{-\nu_1} = \alpha \psi_t X_t H_t^{\alpha-1} \quad (3)$$

- Equilibrium wage condition weights the MPL and outside option of workers and firms:

$$W_t H_t = \eta [\psi_t X_t H_t^\alpha + \kappa_t \theta_t] + (1 - \eta) Z_t$$

where $Z_t = b + \nu_u / \lambda_t^C - \nu_0 \frac{(1 - H_t)^{1-\nu_1}}{1 - \nu_1} / \lambda_t^C$

Demand Shocks Only: Responses to a Demand Shock



Notes: Model with demand shocks only; AIT with $\phi_{\pi}^a = 0.42$. One-standard-deviation shock, starting from steady-state unemployment; mean over 2,000 histories.

Demand Shocks Only: Model-Implied Moments, Monthly Inflation

	Dual mandate	Strict inflation target	
		IT	AIT
Mean:			
U	5.1	5.1	5.1
π^m	2.0	1.8	1.9
R	2.4	2.3	2.4
Standard deviation:			
$\sigma(U)$	0.07	0.07	0.07
$\sigma(\pi^m)$	0.35	0.55	0.36
$\sigma(R)$	0.49	0.68	0.47
Correlations:			
$\text{corr}(U, \pi^m)$	-0.80	-0.86	-0.80
$\text{corr}(g_t, g_{t-1})$	0.28	0.28	0.27

Notes: Model with demand shocks only; AIT with $\phi_\pi^a = 0.42$. π^m : monthly inflation rate, annualized.

$g_t = (U_t - U_{t-1})/U_{t-1}$, quarterly. Model: mean of 500 samples.